ThmSmith Stage -1 proposal: `pid_mediation_proxy` / v1

Sharp joint PNDE/TNIE partial identification under a shared proxy-null bridge fiber

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Locked parameters.

Partial identification tuple. `qid = pid_mediation_proxy`; `specialization = v1`; observable distribution P is the law of $O = (X, A, M, Y, W, Z)$, where $A \in \{0, 1\}$ is a single binary treatment, $M \in \{0, 1\}$ is a mediator, $Y \in [0, 1]$ is an outcome, and $W, Z \in \{0, 1\}$ are negative-control proxies for an unobserved post-treatment mediator–outcome confounder U; parameter $\theta = (\theta_D, \theta_I)$, where $\theta_D = \mathrm{PNDE}(1, 0)$ and $\theta_I = \mathrm{TNIE}(1, 0)$; identifying restrictions are consistency, treatment ignorability conditional on X, mediator/proxy positivity, negative-control exclusion and relevance, and the one-dimensional shared proxy-null bridge-fiber restriction in Assumption 5; the bounding functional is the affine image

$$B(P) = \{(\theta_D^0 + \beta_D \eta, \theta_I^0 + \beta_I \eta) : \eta \in [\ell(P), u(P)]\},$$

where $[\ell(P), u(P)]$ is the observable feasibility interval induced by the rank-deficient bridge equations and the outcome-range constraints.

Abstract

This proposal asks whether negative-control proxies can deliver a sharp *joint* partial-identification statement for the pure natural direct effect and total natural indirect effect when bridge equations are rank deficient. Existing proximal mediation work emphasizes bridge existence and uniqueness for point identification, while mediation-bounds

work warns that separately sharp decompositions need not combine sharply. The proposed kernel is that, with binary proxies and a one-dimensional shared bridge null fiber, the joint identified set for (PNDE, TNIE) is a line segment, not the rectangle obtained by multiplying the two marginal intervals. This would turn weak or redundant negative controls from a nuisance failure of point identification into a testable geometry for mediation decompositions. Sharpness is left as a conjecture; it is not assumed as a bridge realisability condition, but posed as a finite latent-table projection claim. The affine and non-rectangular consequences are routine finite-dimensional algebra once the shared-null setup is fixed.

1 Introduction

Mediation analysis is often used to separate a treatment's effect into a path operating directly on the outcome and a path operating through an intermediate mediator. The decomposition is fragile when a post-treatment variable U confounds the mediator–outcome relation: graphical identification can fail, sensitivity parameters are hard to calibrate, and point-identifying proximal bridge equations can lose uniqueness when the negative controls are weak.

The main proposed finding is Conjecture 1: under a binary-proxy rank-deficient bridge equation whose null direction is shared across the two natural-effect functionals, the sharp joint identified set for (PNDE, TNIE) is the one-dimensional affine image of the observable bridge-feasibility interval. The accompanying algebraic theorem states that the marginal PNDE and TNIE formulas are affine projections of the same coordinate, so the Cartesian product of the two marginal ranges is generally too large if the sharpness conjecture holds.

What is new is the joint-geometry claim. Proximal mediation results such as Dukes et al. (2023) target point identification under bridge conditions; partial-identification mediation results such as Miles et al. (2015) and Miles et al. (2017) give no-proxy pure-direct and natural-indirect bounds; and Gabriel et al. (2023) show that decomposed effect bounds can lose sharpness when combined. Proxy partial-identification results such as Zhang et al. (2023) and Ghassami et al. (2023) show that nonunique bridges can still carry causal information, but they do not formulate a mediation decomposition in which the same bridge-null coordinate moves PNDE and TNIE together. The proposed kernel says that rank-deficient proxies create a *common* latent coordinate that couples the direct and indirect effect intervals, yielding a non-rectangular sharp set with an observable slope.

The proposal sits in the partial-identification cluster: it is closest to Manski/Balke–Pearl style latent-table sharpness, modern proximal partial identification with negative controls, and mediation bounds under unmeasured mediator–outcome confounding. It does not propose a new do-calculus rule or SCM axiom; the SCM and potential-outcome machinery are treated as background language for a finite-dimensional identified-set statement.

2 Related work

Avin et al. (2005) show that path-specific effects, including natural effects, can be unidentified in graphs with intermediate confounding; this is the non-identification regime in which the present proposal works. Cai et al. (2008) derive bounds for direct effects with a confounded intermediate variable; their target is controlled/direct-effect bounds, whereas this proposal targets the joint geometry of natural direct and indirect effects. Imai et al. (2010) frame mediator–outcome confounding through sensitivity analysis, motivating why an observable negative-control coordinate would be useful. Ding and VanderWeele (2016) develop sharp or interpretable sensitivity bounds for mediation effects; this proposal replaces part of the sensitivity region by a proxy-constrained bridge fiber.

Miles et al. (2015) give partial-identification bounds for the pure natural direct effect when cross-world independence and no-exposure-induced confounding assumptions are relaxed; their bounds are componentwise and do not use negative-control bridge equations. Miles et al. (2017) give partial-identification results for natural indirect effects under relaxed mediation assumptions; together with the pure-direct-effect paper, this is the closest no-proxy mediation-bounds thread. Miles et al. (2017) develop semiparametric estimation of a path-specific effect under mediator-outcome confounding, clarifying the adjacent path-specific-effects literature but not supplying a proxy-null joint identified set. Dukes et al. (2023) develop proximal mediation identification using negative controls and bridge functions; the present proposal is about what remains sharply identified when the relevant finite bridge system is not unique. Zhang et al. (2023) show that proximal causal effects can remain point identified without uniqueness of the bridge solution when all admissible bridges agree on the target functional; this proposal instead studies the case where admissible bridges disagree on each mediation component but agree on a one-dimensional joint geometry. Ghassami et al. (2023) develop partial identification of causal effects using proxy variables when proximal point-identification conditions fail; their target is a proxy-bounded causal effect rather than the coupled PNDE/TNIE decomposition line segment proposed here. Miao et al. (2024) study double-negative-control bridge correction in related causal-effect settings; they motivate the rank/completeness issue but do not state a joint PNDE/TNIE partial-identification geometry. Penning de Vries et al. (2023) survey negative-control caveats, including weak or redundant controls; the proposal gives one formal target-loading-specific sense in which redundancy is not all-or-nothing. Gabriel et al. (2023) warn that separately sharp decomposition bounds for mediation effects require careful joint construction rather than naive combination; this is the closest published decomposition-bounds novelty axis, and the proposed line segment is a positive joint sharpness conjecture in a proxy-constrained model.

In-repo, 'ThmSmith/doc/API.md' reports that the ThmSmith ExactID and PartialID catalogues are empty at seed time, so there is no existing ThmSmith proposal whose kernel coincides with this one. The promoted AutoID catalogue in '../doc/API.md' contains partial-identification infrastructure for Manski bounds, Balke–Pearl sharp latent-table bounds, and proximal proxy bounds for ATE-style targets; these are relevant substrate and comparison points, but none records a mediation-specific joint PNDE/TNIE proxy-null identified set.

3 Formal setup

Let $O = (X, A, M, Y, W, Z) \sim P$, with $A, M, W, Z \in \{0, 1\}$ and $Y \in [0, 1]$. The variable U is unobserved and may be affected by A; it may confound the mediator–outcome relation after treatment. The targets are

$$\theta_D = \mathbb{E}\{Y(1, M(0)) - Y(0, M(0))\}, \qquad \theta_I = \mathbb{E}\{Y(1, M(1)) - Y(1, M(0))\}.$$

When a total-effect compatibility check is invoked, write $\theta_T = \mathbb{E}\{Y(1, M(1)) - Y(0, M(0))\}$. For each finite stratum $c = (x, a, m)$, write $r_c(z) = \mathbb{E}[Y \mid X = x, A = a, M = m, Z = z]$, $h_c(w) \in [0, 1]$ for an outcome bridge indexed by $W = w$, and $K_c(z, w) = P(W = w \mid X = x, A = a, M = m, Z = z)$. The observable bridge equation is

$$K_c h_c = r_c.$$

The proposal studies the rank-deficient binary-proxy case where every solution can be written $h_c(\eta) = h_c^0 + \eta n_c$ with a single scalar η shared across the natural-effect functionals. Outcome-range and observable compatibility constraints restrict the scalar to an interval $[\ell(P), u(P)]$. The natural-effect functionals are known affine maps of the bridge family:

$$\theta_D(\eta) = \theta_D^0 + \beta_D\eta, \qquad \theta_I(\eta) = \theta_I^0 + \beta_I\eta,$$

where the constants are computed from P, the mediator distributions $P(M = m \mid A = a, X = x)$, and the chosen bridge representatives $\{h_c^0, n_c\}$. The candidate affine bound is

$$B(P) = \{(\theta_D(\eta), \theta_I(\eta)) : \eta \in [\ell(P), u(P)]\}.$$

The notation $\Theta_D(P)$ and $\Theta_I(P)$ is reserved for the marginal PNDE and TNIE intervals, respectively; only $\Theta_{D,I}(P)$ denotes the sharp joint identified set.

For the sharpness question, let $\mathcal{L}(P)$ denote the finite latent response-type polytope of nonnegative masses over binary response types for $(U, M(0), M(1), W, Z, Y(a, m))$ that reproduce the observable law P and obey the exclusion, consistency, positivity, and shared-null bridge restrictions. For $Q \in \mathcal{L}(P)$, let $\eta(Q)$ be the scalar shared-null coordinate induced by its bridge representative, and define the primitive projection

$$E(P) = \{\eta(Q) : Q \in \mathcal{L}(P)\}.$$

The observable bridge equations imply $E(P) \subseteq [\ell(P), u(P)]$. The nontrivial sharpness question is whether the reverse inclusion holds.

> **Locked parameters repeated.** `qid = pid_mediation_proxy`; `specialization = v1`; observable distribution P is the law of $O = (X, A, M, Y, W, Z)$, where $A \in \{0, 1\}$ is a single binary treatment, $M \in \{0, 1\}$ is a mediator, $Y \in [0, 1]$ is an outcome, and $W, Z \in \{0, 1\}$ are negative-control proxies for an unobserved post-treatment mediator–outcome confounder U; parameter $\theta = (\theta_D, \theta_I)$, where $\theta_D = \mathrm{PNDE}(1, 0)$ and $\theta_I = \mathrm{TNIE}(1, 0)$; identifying restrictions are consistency, treatment ignorability conditional on X, mediator/proxy positivity, negative-control exclusion and relevance, and the one-dimensional shared proxy-null bridge-fiber restriction in Assumption 5; the bounding functional is the affine image
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> $$B(P) = \{(\theta_D^0 + \beta_D\eta, \theta_I^0 + \beta_I\eta) : \eta \in [\ell(P), u(P)]\},$$
>
> where $[\ell(P), u(P)]$ is the observable feasibility interval induced by the rank-deficient bridge equations and the outcome-range constraints.

4 Assumptions

Assumption 1 (Finite binary mediation system)**.** The variables satisfy $A, M, W, Z \in \{0, 1\}$, $Y \in [0, 1]$, and X has finite support. All cells used in conditional probabilities below have strictly positive probability.

Assumption 2 (Consistency and treatment ignorability)**.** Observed M and Y equal their corresponding factual potential outcomes, and A is conditionally independent of the relevant treatment-indexed potential outcomes and proxy response types given X.

Assumption 3 (Negative-control exclusion)**.** Conditional on (X, A, M, U), Z carries no direct information about the outcome bridge other than through U, and W is an outcome-inducing proxy for U with no direct treatment or mediator effect except through the bridge relation encoded by $K_c h_c = r_c$.

Assumption 4 (Rank-deficient binary bridge)**.** For each $c = (x, a, m)$, the observed 2×2 matrix $K_c(z, w) = P(W = w \mid X = x, A = a, M = m, Z = z)$ has rank one, the equation $K_c h_c = r_c$ is feasible, and its solution set is $\{h_c^0 + \eta_c n_c : \eta_c \in I_c(P)\}$ for a nonzero null vector n_c.

Assumption 5 (Shared proxy-null coordinate)**.** The feasible bridge solutions relevant to θ_D and θ_I are indexed by a single scalar η, so $\eta_c = s_c\eta + t_c$ for known observable signs/scales s_c, t_c. The common feasible interval $[\ell(P), u(P)]$ is the intersection of the stratum intervals after this reparameterisation.

Assumption 6 (Affine natural-effect loading)**.** The PNDE and TNIE functionals induced by the bridge family are $\theta_D(\eta) = \theta_D^0 + \beta_D\eta$ and $\theta_I(\eta) = \theta_I^0 + \beta_I\eta$, with $(\beta_D, \beta_I) \neq (0, 0)$. The constants $\theta_D^0, \theta_I^0, \beta_D, \beta_I, \ell(P), u(P)$ are observable functionals of P under Assumptions 1–5.

Assumption 7 (Primitive latent-table feasibility)**.** The finite response-type polytope $\mathcal{L}(P)$ defined in the formal setup is nonempty. Every $Q \in \mathcal{L}(P)$ reproduces P, satisfies Assumptions 1–6, and induces a well-defined shared-null scalar $\eta(Q) \in [\ell(P), u(P)]$. This assumption does not assert that every $\eta \in [\ell(P), u(P)]$ is attainable.

Assumption 8 (Total-effect bridge compatibility)**.** The same bridge family that induces $\theta_D(\eta)$ and $\theta_I(\eta)$ also identifies the total effect θ_T from P, so the total-effect functional has no residual loading on the shared proxy-null coordinate.

5 Main results

Theorem 1 (Theorem 1: affine marginal proxy-null intervals)**.** *Under Assumptions 1–6, the candidate marginal ranges generated by the shared bridge fiber are*

$$\Theta_D(P) = [\min_{\eta \in [\ell(P), u(P)]} \theta_D(\eta), \max_{\eta \in [\ell(P), u(P)]} \theta_D(\eta)]$$

and

$$\Theta_I(P) = [\min_{\eta \in [\ell(P), u(P)]} \theta_I(\eta), \max_{\eta \in [\ell(P), u(P)]} \theta_I(\eta)].$$

If both β_D and β_I are nonzero and $\ell(P) < u(P)$, then the affine image $B(P)$ is a line segment and is a strict subset of $\Theta_D(P) \times \Theta_I(P)$.

Why this is new and worth studying. The closest published warning is Gabriel et al. (2023), which shows that separately sharp decomposed bounds need not combine sharply; this theorem identifies a concrete proxy-null mechanism that explains why the product of candidate marginal PNDE and TNIE intervals is too large. It advances the field by giving a finite-dimensional diagnostic: compute the two loadings β_D, β_I and the shared feasibility interval before reporting separate mediation bounds as if they were jointly attainable.

Conjecture 1 (Conjecture 1: sharp joint proxy-null mediation set (NOT YET PROVED))**.** *Under Assumptions 1–6 and Assumption 7, the primitive latent-table projection satisfies $E(P) = [\ell(P), u(P)]$. Consequently, the sharp joint identified set for* $(\mathrm{PNDE}(1,0), \mathrm{TNIE}(1,0))$ *is exactly*

$$\Theta_{D,I}(P) = B(P) = \{(\theta_D^0 + \beta_D \eta, \theta_I^0 + \beta_I \eta) : \eta \in [\ell(P), u(P)]\}.$$

Equivalently, every point on the line segment is attainable by a model with observable law P, and no point off the segment is attainable.

Why this is new and worth studying. Dukes et al. (2023) focus on proximal mediation identification when bridge conditions are strong enough for point identification, while Miles et al. (2015), Miles et al. (2017), and Ding and VanderWeele (2016) bound natural effects without exploiting a proxy-null bridge fiber. Zhang et al. (2023) and Ghassami et al. (2023) are the closest proxy-based nonuniqueness and partial-identification comparators, but they treat target functionals whose bridge-null disagreement is either eliminated or bounded directly rather than a pair of natural-effect components that share the same null coordinate. The conjecture is not a weakerassumption variant of those papers or of the no-proxy pure-direct/natural-indirect bounds: it asserts that primitive latent-table feasibility fills the whole observable proxy-null interval and therefore creates a new sharp joint geometry in mediation. Resolving it would make weak negative controls operational, because the remaining unidentified direction would be an observable one-dimensional sensitivity coordinate rather than an analyst-chosen sensitivity parameter.

Conjecture 2 (Conjecture 2: total-effect hyperplane compatibility (NOT YET PROVED))**.** *Under Assumptions 1–6, Assumption 7, and Assumption 8,*

$$\beta_D + \beta_I = 0 \quad \textit{and} \quad \theta_D(\eta) + \theta_I(\eta) = \theta_T \quad \textit{for all } \eta \in [\ell(P), u(P)].$$

Why this is new and worth studying. The closest decomposition literature treats the direct and indirect pieces as separate bounded targets and then asks how they add up. This conjecture gives a proxy-null falsification check: if an estimated shared-null model produces a nonzero total-effect loading, the locked assumptions are incoherent. The result would connect proximal mediation bounds to the familiar natural-effect decomposition while preserving the non-rectangular joint set.

6 Sanity-check corollaries

Corollary 1 (Full-rank proxy reduction)**.** *If Assumption 4 is replaced by full column rank of every K_c, then $\ell(P) = u(P)$, so $B(P)$ collapses to a singleton. This is the proximal point-identification corner case studied by Dukes et al. (2023).*

Corollary 2 (Total-effect sanity check)**.** *Under Assumption 8 and the conclusion of Conjecture 2, the line segment $B(P)$ lies in the observable affine hyperplane $\theta_D + \theta_I = \theta_T$. Thus the proposal cannot create spurious variation in the total effect while varying the direct and indirect pieces.*

7 Proof-sketch route

The algebraic theorem reduces to two facts from finite-dimensional linear programming: a rank-one 2×2 bridge equation has a one-dimensional affine solution fiber, and a linear functional maps a compact interval to a compact interval whose endpoints occur at the scalar endpoints. The sharpness conjecture should be attacked by a finite latent-type construction: enumerate binary response types for $(M(0), M(1), W, Z, Y(a, m))$, impose the observed cell-probability and bridge equations as linear constraints, and show that the resulting polytope projection $E(P)$ equals the whole observable interval $[\ell(P), u(P)]$. Equivalently, fixing $\eta \in [\ell(P), u(P)]$ should admit a nonnegative latent table that reproduces P and attains $(\theta_D(\eta), \theta_I(\eta))$. The boundary cases needing explicit algebra are the two endpoint tables $\eta = \ell(P)$ and $\eta = u(P)$, plus the degeneracies $\beta_D = 0$, $\beta_I = 0$, and $\ell(P) = u(P)$.

8 Honest scope flags

Theorem 1 is a routine algebra statement conditional on the locked shared-null setup. Conjectures 1 and 2 are not yet proved; their hard part is the latent-table sharpness construction, specifically the equality $E(P) = [\ell(P), u(P)]$, not the affine projection algebra. The strongest scope restriction is Assumption 5: if the PNDE and TNIE depend on independent bridge-null coordinates, the joint set can become higher-dimensional and the proposed line segment is false. This version also withdraws the v1–v2 formulation that assumed endpoint bridge realisability; attainability is now the open content of Conjecture 1. This angle withdraws any claim that proxy rank deficiency by itself point-identifies a single natural effect; the kernel is joint geometry under a shared null coordinate.

9 Iteration trail

- v1 (pivot) — initial draft for angle 2: joint PNDE/TNIE sharp set as a shared proxy-null line segment; prior verdict: none.
- v2 (revise) — corrected the Miles et al. bibliography title, reserved $\Theta_I(P)$ for the TNIE marginal interval while naming the joint set $\Theta_{D,I}(P)$, and added the missing Zhang et al. and Ghassami et al. proxy partial-identification comparisons; prior verdict: REVISE.
- v3 (revise) — replaced sharp bridge realisability by primitive latent-table feasibility, made endpoint/interior attainability the explicit content of Conjecture 1, and added the missing pure direct-effect and path-specific-effect mediation comparisons; prior verdict: REVISE.

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